

Лабораторная работа №4

Дисциплина: Основы информационной безопасности

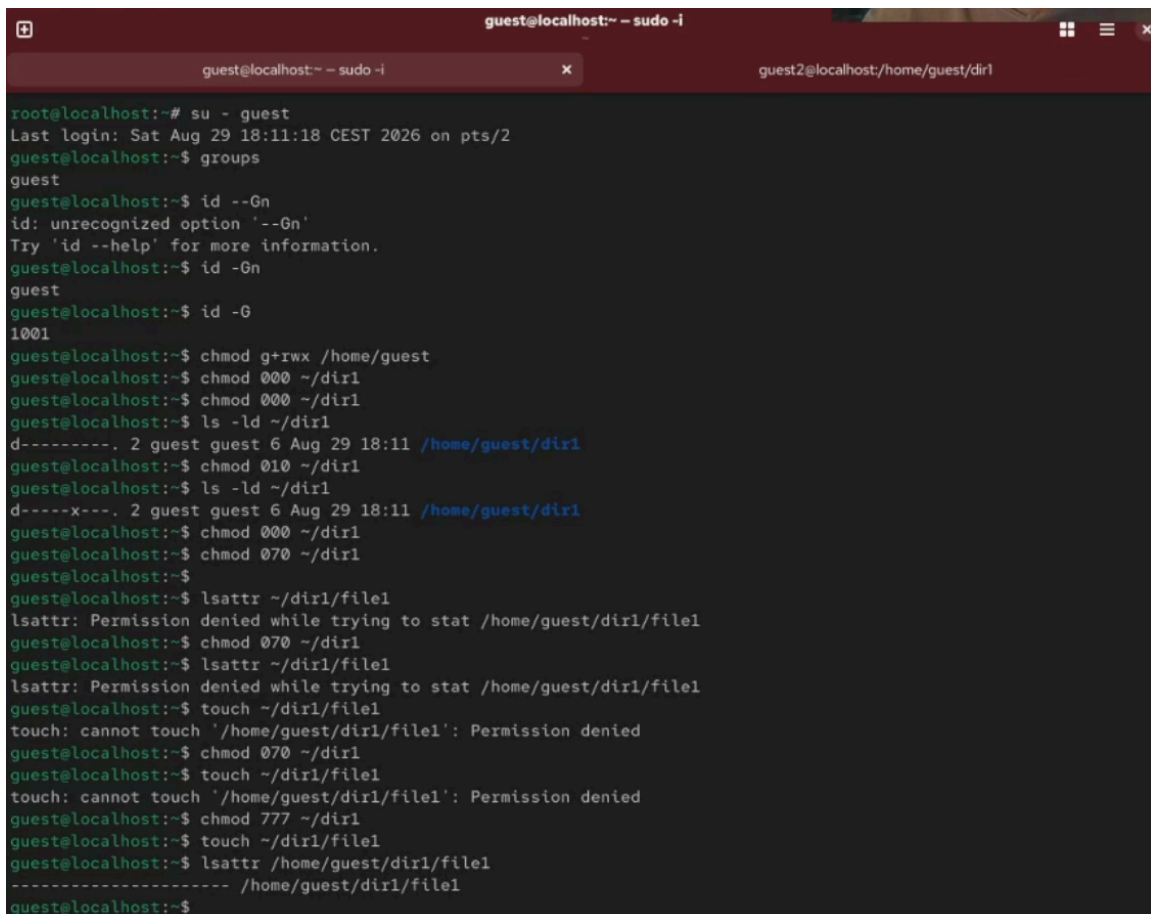
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Цель работы

Получить практические навыки по работе в консоли с расширенными атрибутами файлов.

Выполнение лабораторной работы

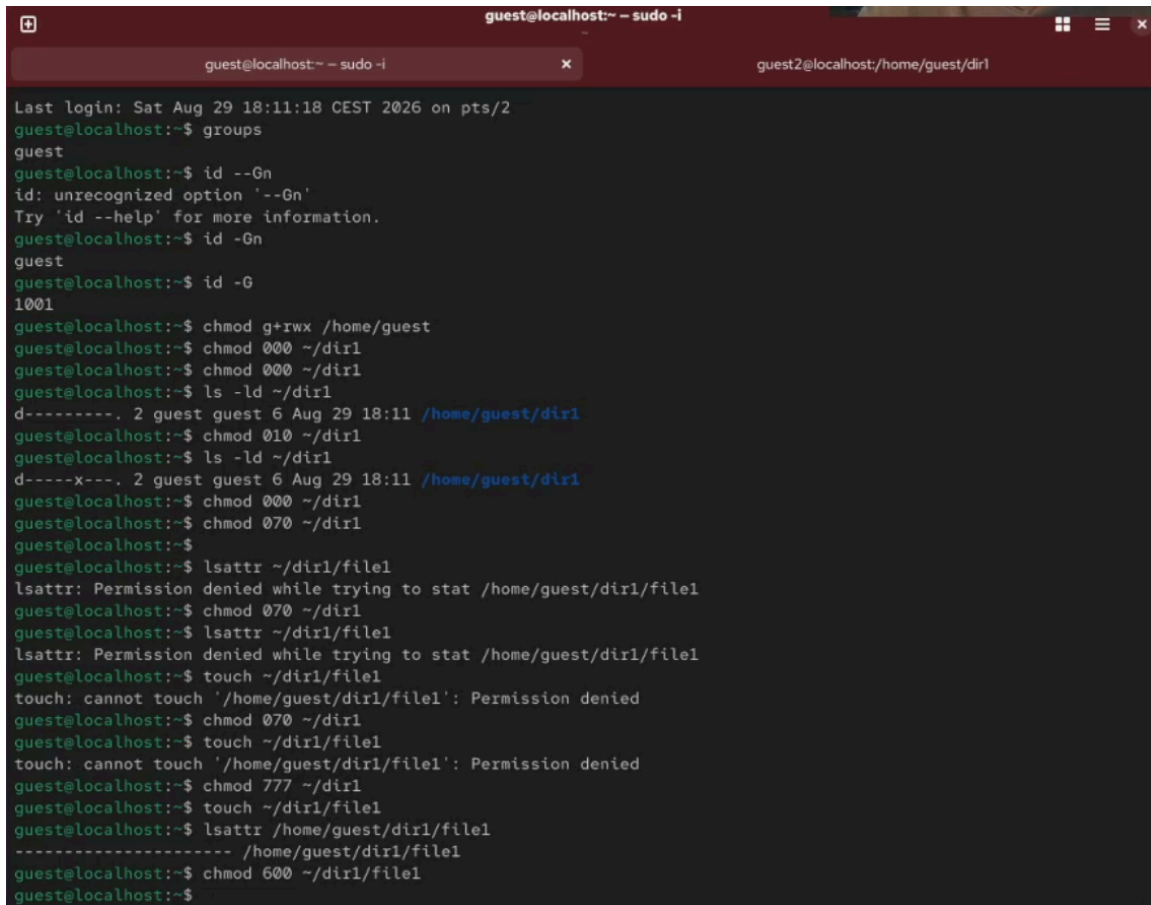
От имени пользователя geust определяю расширенные атрибуты файла dir1/file1.



```
guest@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
root@localhost:~# su - guest
Last login: Sat Aug 29 18:11:18 CEST 2026 on pts/2
guest@localhost:~$ groups
guest
guest@localhost:~$ id --Gn
id: unrecognized option '--Gn'
Try 'id --help' for more information.
guest@localhost:~$ id -Gn
guest
guest@localhost:~$ id -G
1001
guest@localhost:~$ chmod g+rxw /home/guest
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ ls -ld ~/dir1
d------. 2 guest guest 6 Aug 29 18:11 /home/guest/dir1
guest@localhost:~$ chmod 010 ~/dir1
guest@localhost:~$ ls -ld ~/dir1
d-----x---. 2 guest guest 6 Aug 29 18:11 /home/guest/dir1
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$
guest@localhost:~$ lsattr ~/dir1/file1
lsattr: Permission denied while trying to stat /home/guest/dir1/file1
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ lsattr ~/dir1/file1
lsattr: Permission denied while trying to stat /home/guest/dir1/file1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 777 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
----- /home/guest/dir1/file1
guest@localhost:~$
```

Рисунок 1. атрибуты file1

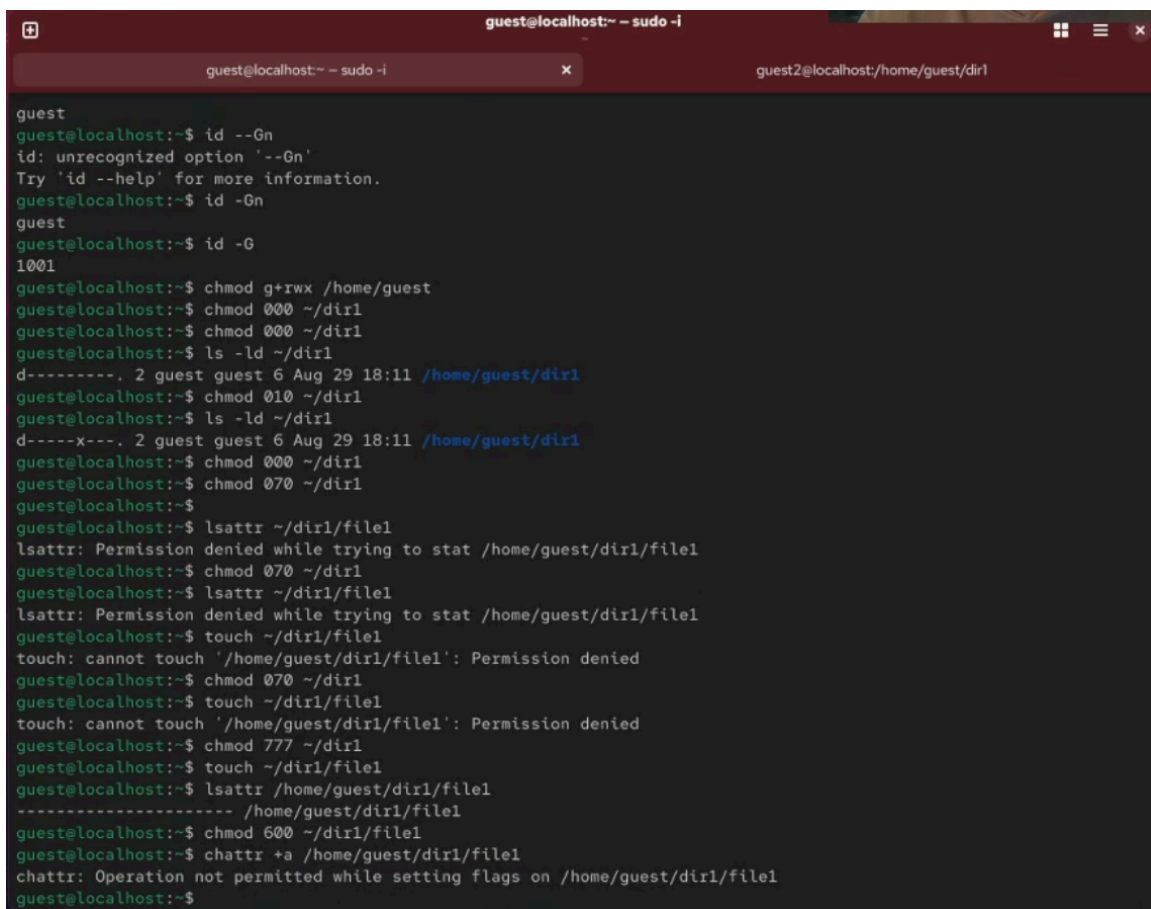
Изменяю права доступа для файла с помощью chmod 600.



```
guest@localhost:~ - sudo -i
Last login: Sat Aug 29 18:11:18 CEST 2026 on pts/2
guest@localhost:~$ groups
guest
guest@localhost:~$ id --Gn
id: unrecognized option '--Gn'
Try 'id --help' for more information.
guest@localhost:~$ id -G
guest
guest@localhost:~$ id -G
1001
guest@localhost:~$ chmod g+rxw /home/guest
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ ls -ld ~/dir1
d------. 2 guest guest 6 Aug 29 18:11 /home/guest/dir1
guest@localhost:~$ chmod 010 ~/dir1
guest@localhost:~$ ls -ld ~/dir1
d----x---. 2 guest guest 6 Aug 29 18:11 /home/guest/dir1
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$
guest@localhost:~$ lsattr ~/dir1/file1
lsattr: Permission denied while trying to stat /home/guest/dir1/file1
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ lsattr ~/dir1/file1
lsattr: Permission denied while trying to stat /home/guest/dir1/file1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 777 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
----- /home/guest/dir1/file1
guest@localhost:~$ chmod 600 ~/dir1/file1
guest@localhost:~$
```

Рисунок 2. chmod 600

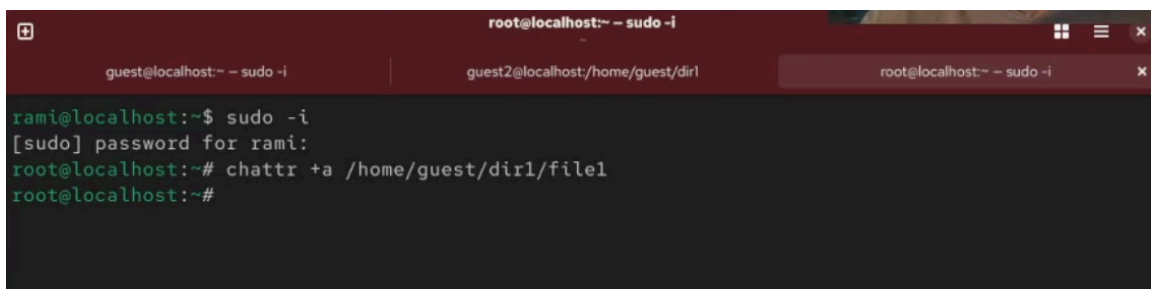
Устанавливаю расширенный атрибут от имени пользователя guest и получаю отказ действия в ответе.



```
guest@localhost:~$ id --Gn
id: unrecognized option '--Gn'
Try 'id --help' for more information.
guest@localhost:~$ id -G
guest
guest@localhost:~$ id -G
1001
guest@localhost:~$ chmod g+rx /home/guest
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ ls -ld ~/dir1
d-----x. 2 guest guest 6 Aug 29 18:11 /home/guest/dir1
guest@localhost:~$ chmod 010 ~/dir1
guest@localhost:~$ ls -ld ~/dir1
d-----x. 2 guest guest 6 Aug 29 18:11 /home/guest/dir1
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$
guest@localhost:~$ lsattr ~/dir1/file1
lsattr: Permission denied while trying to stat /home/guest/dir1/file1
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ lsattr ~/dir1/file1
lsattr: Permission denied while trying to stat /home/guest/dir1/file1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 777 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
----- /home/guest/dir1/file1
guest@localhost:~$ chmod 600 ~/dir1/file1
guest@localhost:~$ chattr +a /home/guest/dir1/file1
chattr: Operation not permitted while setting flags on /home/guest/dir1/file1
guest@localhost:~$
```

Рисунок 3. Устаноавка атрибута без разрешения администратора

Устанавливаю права от имени суперпользователя и получается установить атрибут и проверяю работу с помощью lsattr.



```
rami@localhost:~$ sudo -i
[sudo] password for rami:
root@localhost:~# chattr +a /home/guest/dir1/file1
root@localhost:~#
```

Рисунок 4. Устаноавка атрибута с разрешением администратора

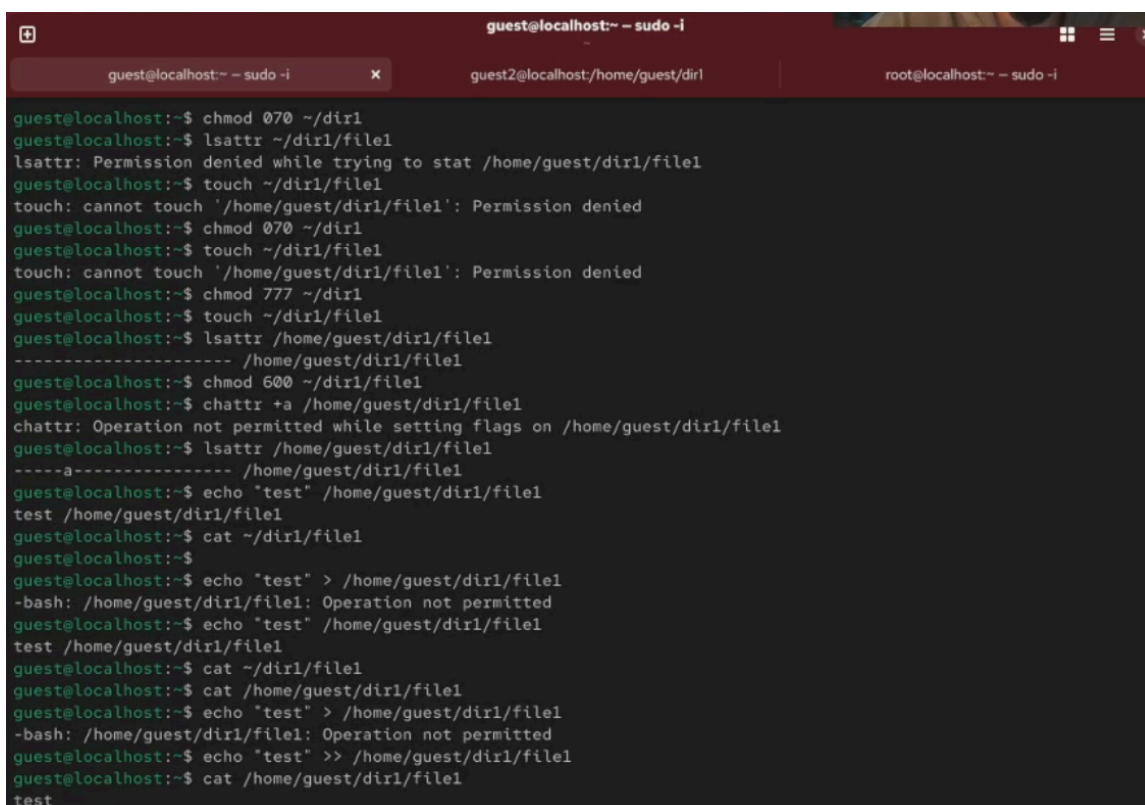
От пользователя guest проверяю правильность установления атрибута

```
guest@localhost:~ -- sudo -i
guest@localhost:~ -- sudo -i x
guest2@localhost:/home/guest/dir1
root@localhost:~ -- sudo -i

id: unrecognized option '--Gn'
Try 'id --help' for more information.
guest@localhost:~$ id -Gn
guest
guest@localhost:~$ id -G
1001
guest@localhost:~$ chmod g+rxw /home/guest
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ ls -ld ~/dir1
d------. 2 guest guest 6 Aug 29 18:11 /home/guest/dir1
guest@localhost:~$ chmod 010 ~/dir1
guest@localhost:~$ ls -ld ~/dir1
d-----x---. 2 guest guest 6 Aug 29 18:11 /home/guest/dir1
guest@localhost:~$ chmod 000 ~/dir1
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ lsattr ~/dir1/file1
lsattr: Permission denied while trying to stat /home/guest/dir1/file1
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ lsattr ~/dir1/file1
lsattr: Permission denied while trying to stat /home/guest/dir1/file1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 777 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
-----a----- /home/guest/dir1/file1
guest@localhost:~$ chmod 600 ~/dir1/file1
guest@localhost:~$ chattr +a /home/guest/dir1/file1
chattr: Operation not permitted while setting flags on /home/guest/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
-----a----- /home/guest/dir1/file1
guest@localhost:~$
```

Рисунок 5. проверка атрибута командной: lsattr /home/guest/dir1/file1

Выполняю дозапись в файл с помощью echo и проверяю командой cat.

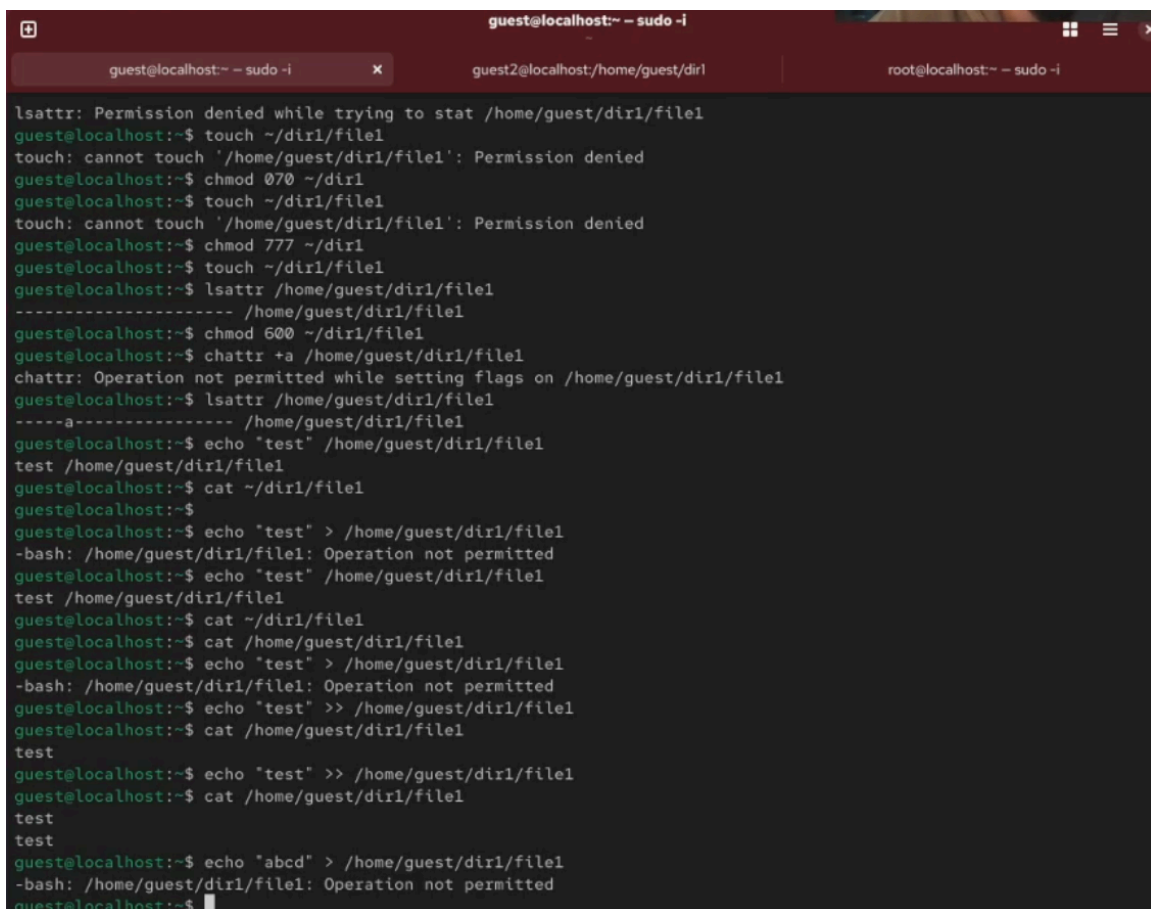


```
guest@localhost:~ - sudo -i
guest2@localhost/home/guest/dir1
root@localhost:~ - sudo -i

guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ lsattr ~/dir1/file1
lsattr: Permission denied while trying to stat /home/guest/dir1/file1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 777 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
----- /home/guest/dir1/file1
guest@localhost:~$ chmod 600 ~/dir1/file1
guest@localhost:~$ chattr +a /home/guest/dir1/file1
chattr: Operation not permitted while setting flags on /home/guest/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
----a----- /home/guest/dir1/file1
guest@localhost:~$ echo "test" /home/guest/dir1/file1
test /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
test
guest@localhost:~$
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" /home/guest/dir1/file1
test /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
test
guest@localhost:~$ cat /home/guest/dir1/file1
test
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
test
guest@localhost:~$ cat /home/guest/dir1/file1
test
test
```

Рисунок 6. Дозапись в file1

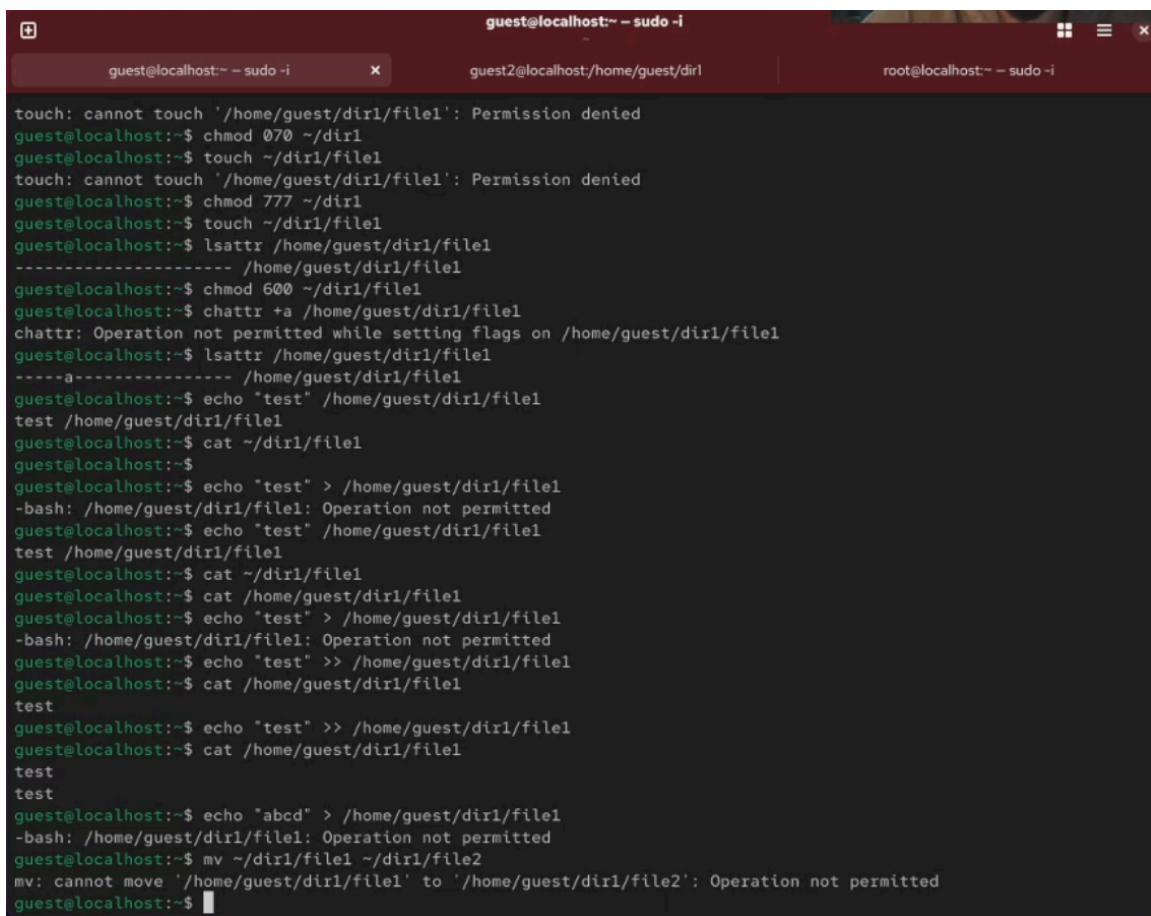
Пробую стереть имеющуюся в file1 информацию и получаю отказ в ответ.



```
lsattr: Permission denied while trying to stat /home/guest/dirl/file1
guest@localhost:~$ touch ~/dirl/file1
touch: cannot touch '/home/guest/dirl/file1': Permission denied
guest@localhost:~$ chmod 070 ~/dirl
guest@localhost:~$ touch ~/dirl/file1
touch: cannot touch '/home/guest/dirl/file1': Permission denied
guest@localhost:~$ chmod 777 ~/dirl
guest@localhost:~$ touch ~/dirl/file1
guest@localhost:~$ lsattr /home/guest/dirl/file1
----- /home/guest/dirl/file1
guest@localhost:~$ chmod 600 ~/dirl/file1
guest@localhost:~$ chattr +a /home/guest/dirl/file1
chattr: Operation not permitted while setting flags on /home/guest/dirl/file1
guest@localhost:~$ lsattr /home/guest/dirl/file1
----a----- /home/guest/dirl/file1
guest@localhost:~$ echo "test" /home/guest/dirl/file1
test /home/guest/dirl/file1
guest@localhost:~$ cat ~/dirl/file1
guest@localhost:~$
guest@localhost:~$ echo "test" > /home/guest/dirl/file1
-bash: /home/guest/dirl/file1: Operation not permitted
guest@localhost:~$ echo "test" /home/guest/dirl/file1
test /home/guest/dirl/file1
guest@localhost:~$ cat ~/dirl/file1
guest@localhost:~$ cat /home/guest/dirl/file1
guest@localhost:~$ echo "test" > /home/guest/dirl/file1
-bash: /home/guest/dirl/file1: Operation not permitted
guest@localhost:~$ echo "test" >> /home/guest/dirl/file1
guest@localhost:~$ cat /home/guest/dirl/file1
test
guest@localhost:~$ echo "test" >> /home/guest/dirl/file1
guest@localhost:~$ cat /home/guest/dirl/file1
test
test
guest@localhost:~$ echo "abcd" > /home/guest/dirl/file1
-bash: /home/guest/dirl/file1: Operation not permitted
guest@localhost:~$
```

Рисунок 7. команда echo «abcd» > /home/guest/dirl/file1

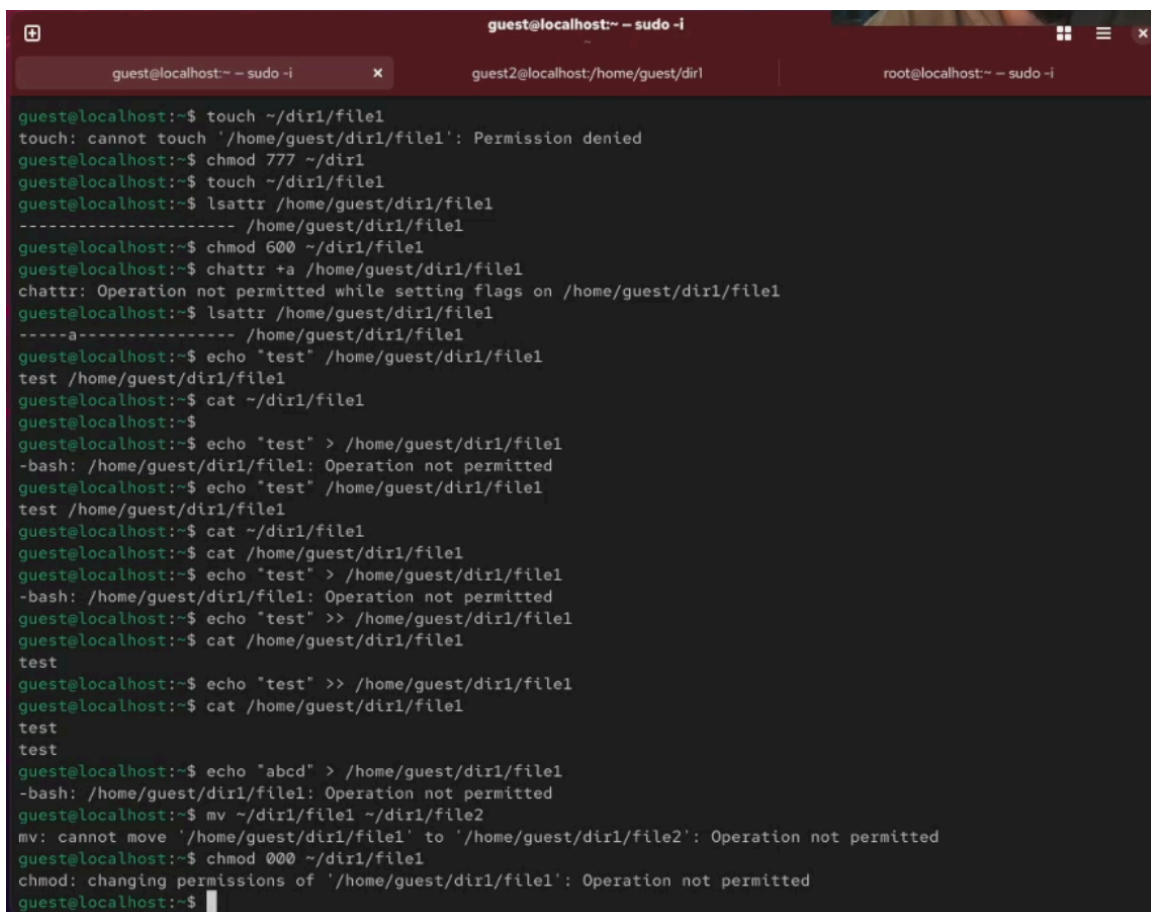
Получаю отказ при попытке переименовать файл.



```
guest@localhost:~ - sudo -i
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 070 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 777 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
----- /home/guest/dir1/file1
guest@localhost:~$ chmod 600 ~/dir1/file1
guest@localhost:~$ chattr +a /home/guest/dir1/file1
chattr: Operation not permitted while setting flags on /home/guest/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
-----a----- /home/guest/dir1/file1
guest@localhost:~$ echo "test" /home/guest/dir1/file1
test /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
guest@localhost:~$
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" /home/guest/dir1/file1
test /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
test
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
test
test
guest@localhost:~$ echo "abcd" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted
guest@localhost:~$
```

Рисунок 8. Попытка переименовать файл

Когда пытаюсь изменить права доступа также получаю отказ.

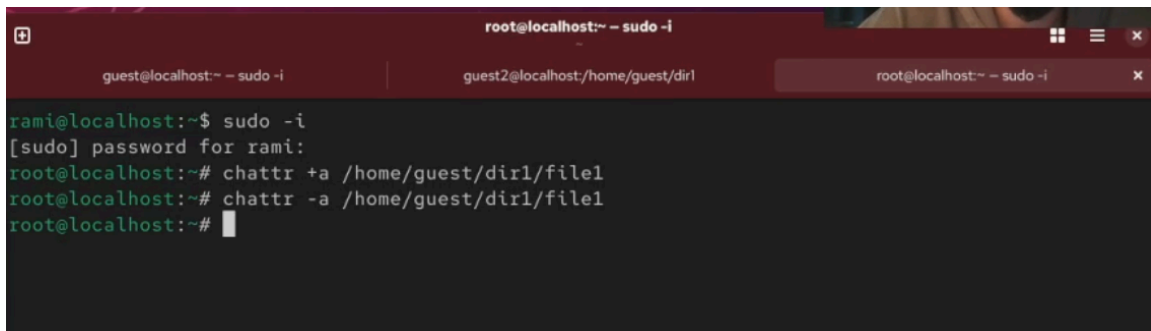


```
guest@localhost:~ - sudo -i
guest2@localhost/home/guest/dir1
root@localhost:~ - sudo -i

guest@localhost:~$ touch ~/dir1/file1
touch: cannot touch '/home/guest/dir1/file1': Permission denied
guest@localhost:~$ chmod 777 ~/dir1
guest@localhost:~$ touch ~/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
----- /home/guest/dir1/file1
guest@localhost:~$ chmod 600 ~/dir1/file1
guest@localhost:~$ chattr +a /home/guest/dir1/file1
chattr: Operation not permitted while setting flags on /home/guest/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
-----a----- /home/guest/dir1/file1
guest@localhost:~$ echo "test" /home/guest/dir1/file1
test /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
guest@localhost:~$
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" /home/guest/dir1/file1
test /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
test
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
test
test
guest@localhost:~$ echo "abcd" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted
guest@localhost:~$ chmod 000 ~/dir1/file1
chmod: changing permissions of '/home/guest/dir1/file1': Operation not permitted
guest@localhost:~$
```

Рисунок 9. Изменение права доступа

Снимаю расширенный атрибут с файла.

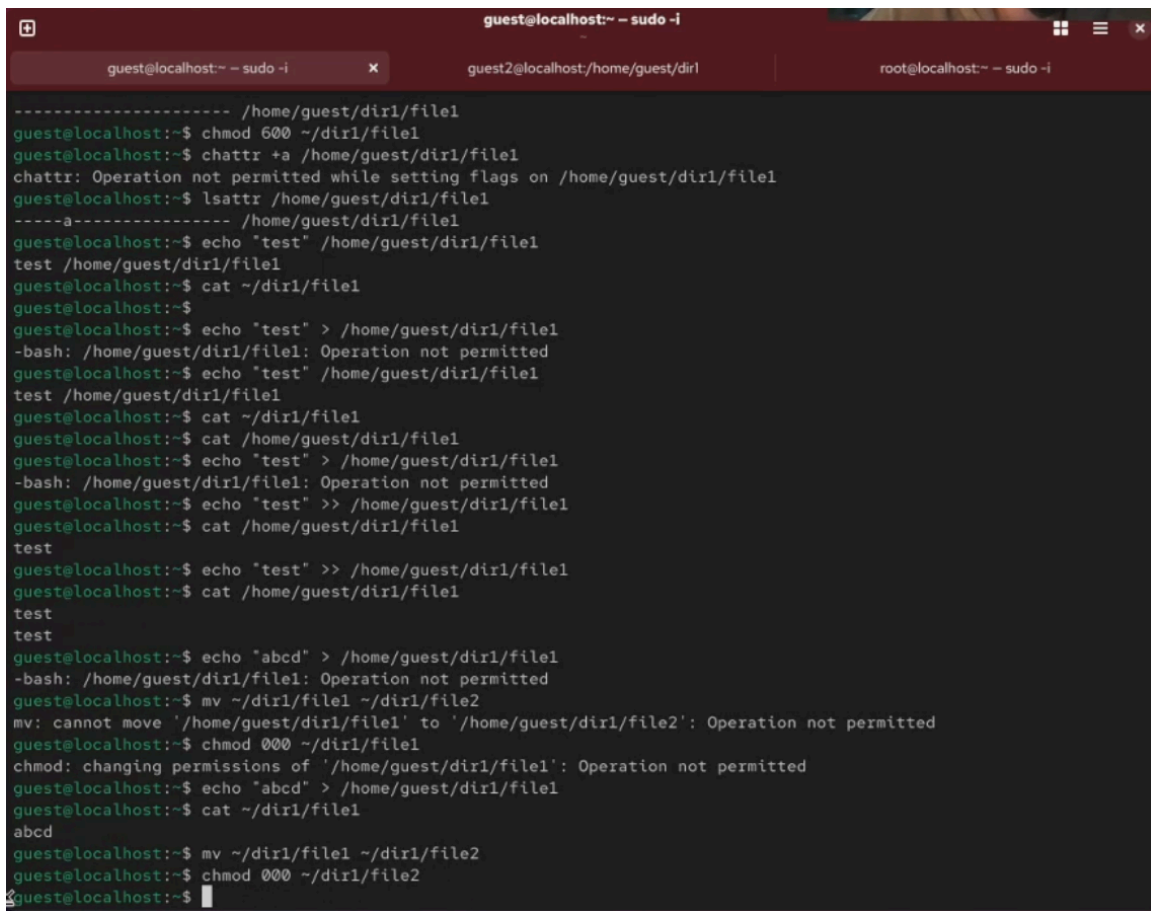


```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost/home/guest/dir1
root@localhost:~ - sudo -i

rami@localhost:~$ sudo -i
[sudo] password for rami:
root@localhost:~# chattr +a /home/guest/dir1/file1
root@localhost:~# chattr -a /home/guest/dir1/file1
root@localhost:~#
```

Рисунок 10. Снятие атрибут а

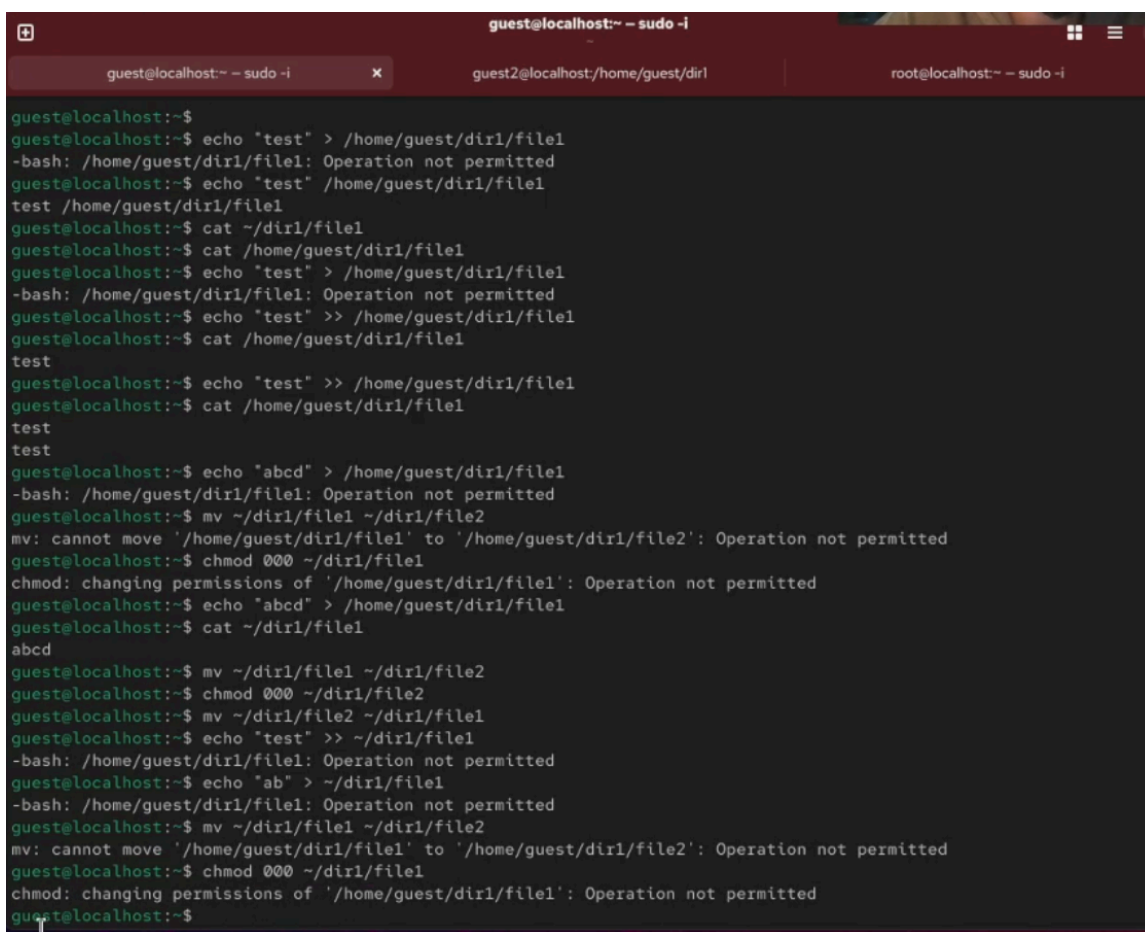
Проверяю файл на чтение, переименование, изменение прав доступа и выполнение.



```
----- /home/guest/dir1/file1
guest@localhost:~$ chmod 600 ~/dir1/file1
guest@localhost:~$ chattr +a /home/guest/dir1/file1
chattr: Operation not permitted while setting flags on /home/guest/dir1/file1
guest@localhost:~$ lsattr /home/guest/dir1/file1
-----a----- /home/guest/dir1/file1
guest@localhost:~$ echo "test" /home/guest/dir1/file1
test /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
guest@localhost:~$
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" /home/guest/dir1/file1
test /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
test
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
test
test
guest@localhost:~$ echo "abcd" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted
guest@localhost:~$ chmod 000 ~/dir1/file1
chmod: changing permissions of '/home/guest/dir1/file1': Operation not permitted
guest@localhost:~$ echo "abcd" > /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
abcd
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
guest@localhost:~$ chmod 000 ~/dir1/file2
guest@localhost:~$
```

Рисунок 11. Проверка файла

Далее повторяю все проделанные действия но с расширенный атрибут i.



```
guest@localhost:~$  
guest@localhost:~$ echo "test" > /home/guest/dir1/file1  
-bash: /home/guest/dir1/file1: Operation not permitted  
guest@localhost:~$ echo "test" /home/guest/dir1/file1  
test /home/guest/dir1/file1  
guest@localhost:~$ cat ~/dir1/file1  
guest@localhost:~$ cat /home/guest/dir1/file1  
guest@localhost:~$ echo "test" > /home/guest/dir1/file1  
-bash: /home/guest/dir1/file1: Operation not permitted  
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1  
guest@localhost:~$ cat /home/guest/dir1/file1  
test  
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1  
guest@localhost:~$ cat /home/guest/dir1/file1  
test  
test  
guest@localhost:~$ echo "abcd" > /home/guest/dir1/file1  
-bash: /home/guest/dir1/file1: Operation not permitted  
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2  
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted  
guest@localhost:~$ chmod 000 ~/dir1/file1  
chmod: changing permissions of '/home/guest/dir1/file1': Operation not permitted  
guest@localhost:~$ echo "abcd" > /home/guest/dir1/file1  
guest@localhost:~$ cat ~/dir1/file1  
abcd  
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2  
guest@localhost:~$ chmod 000 ~/dir1/file2  
guest@localhost:~$ mv ~/dir1/file2 ~/dir1/file1  
guest@localhost:~$ echo "test" >> ~/dir1/file1  
-bash: /home/guest/dir1/file1: Operation not permitted  
guest@localhost:~$ echo "ab" > ~/dir1/file1  
-bash: /home/guest/dir1/file1: Operation not permitted  
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2  
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted  
guest@localhost:~$ chmod 000 ~/dir1/file1  
chmod: changing permissions of '/home/guest/dir1/file1': Operation not permitted  
guest@localhost:~$
```

Рисунок 12. Название рисунка

Выводы

В результате выполнения работы мы повысили свои навыки использования интерфейса командой строки (CLI), познакомились на примерах с тем, как используются основные и расширенные атрибуты при разграничении доступа. Имели возможность связать теорию дискреционного разделения доступа (дискреционная политика безопасности) с её реализацией на практике в ОС Linux. Составили наглядные таблицы, поясняющие какие операции возможны при тех или иных установленных правах. Опробовали действие на практике расширенных атрибутов «a» и «i».

Список литературы

https://esystem.rudn.ru/pluginfile.php/3096711/mod_resource/content/3/004-lab_discret_extattr.pdf